Tutorial 5

Prepare a test plan and an online questionnaire

Test scenario

You are working as a UX researcher and you have been assigned to conduct usability testing for Snapfish.

https://www.snapfish.co.uk/home

A large number of Snapfish clients are parents. To retain and expand on this type of clientele (which is responsible for considerable profits), Snapfish has asked you to conduct a study with users and find out about the ease of use of the feature of their website to create a bespoke photo book.

Work in pairs or in groups of a maximum 3 people and explore for 5-10 minutes Snapfish. Identify a couple of tasks you would like to investigate to address usability issues. Before you organise a study you need to have a clear understanding of the tasks you want to study related to user behaviour and preference. For example:

- if it provides the required tools to complete tasks (e.g. to create a photobook);
- if the design of the application conveys the available functionality (to create a photo book, to upload images, etc.);
- if it provides enough tools to modify layouts/templates/designs;
- the average time it takes to complete specific tasks;
- is there enough help provided to overcome difficulties;
- if there is there enough documentation;
- how users feel about the overall usability of the website.

You need to collect performance and preference data related to the main user experience metrics.

Organise your study around a scenario where you ask the participants to create a 8X8 inches bespoke photo book album with a birthday theme. They need to include at least 10 pictures and design the cover page, the back page, and a few pages inside the photo book (at least 6). Each page of the photo book should have a different style. The photo book layout should be suitable for the pictures to be included. Ask the participants to use short descriptive text on the photo book pages (in each picture or a group of pictures per page). Ask them to customize the cover page, the background colour of the photo book pages, borders, and embellishments.

Plan your test for an **online**, **unmoderated** usability testing session.

Task 1. As a group complete all of the following tasks:

- Form a couple of research questions around which you will form your study;

- Form each research question with a couple of hypotheses to serve descriptive, experimental and correlational research;
- Define the dependent and independent variables for each hypothesis you are evaluating;
- Express the hypotheses you formed mathematically.
- Identify what data you need to collect to evaluate your hypotheses.

Tips on Research Questions – What are we trying to find out based on the specific study?

1) What is the average time participants took to complete a specific task? (**Descriptive**)
2) Does having previously uploaded pictures on the site affect the average time required to complete a photobook? (**Experimental**)
3) Does prior experience in creating a photobook affect the average time required to complete a photobook? (**Correlational**)

Model answer:

Research Question A (Efficiency & Success)

RQ-A: How efficiently and successfully can target users create an 8×8″ birthday photo book on Snapfish?

A1. Descriptive hypothesis

- ***H_0 (desc):*** *Median time-on-task for the full creation flow ≥ 10 minutes; completion rate ≤ 70%.*
- ***H_1 (desc):*** *Median time-on-task < 10 minutes; completion rate > 70%.*
- **Dependent Variables (DVs):** time-on-task T (sec), task completion $C \in \{0,1\}$
- **Independent Variables (IVs):** none (descriptive/baseline).
- **Math:**
  $T<600$ and $E[C]>0.70$
- **Data to collect:** per-participant start/end timestamps, success flag, abandon reason; screen/path logs.

A2. Experimental hypothesis (UI assist vs. no assist)

- **Condition:**
  - **Guided start:** pre-filtered to 8×8 birthday book + inline tips
  - **Standard flow:** normal site experience
- ***H_0 (null):*** *There is no difference in efficiency and success outcomes between the **Guided start** condition and the **Standard flow** condition.*
- ***H_1 (alternative):*** *There is a difference in efficiency and/or success outcomes between the **Guided start** condition and the **Standard flow** condition.*

- **Outcomes (DVs): efficiency and success — i.e., time-on-task *T*, error count *E*, completion *C* (success/fail), and SEQ ease score *S* (1–7).**
- **Independent Variables (IVs):** interface variant V∈{guided,standard}
- **Mathematical (two-tailed) equivalents**
    - **Let V∈{guided,standard}**
    - **Time: μTguided=μTstandard**
    - **Errors: μEguided=μEstandard**
    - **Completion: p(C=1|guided)=p(C=1|standard)**
    - **SEQ: μSguided=μSstandard\mu_S**
- **Data to collect:** variant assignment, timestamps, error/misclick counts, SEQ after core tasks, completion flag.

A3. Correlational hypothesis (experience ↔ performance)

- ***H_0 (corr):*** *Prior photobook experience is not associated with efficiency or success.*
- ***H_1 (corr):*** *More experience → faster time and higher success/SEQ.*
- **Dependent Variables (DVs):** *T,C,S.*
- **IV:** self-reported experience level X (e.g., 0–3 scale).
- **Math:**
  ρX,T<0, ρX,T>0, logit p(C=1)=β0+β1X, β1>0 *[do not worry if you do not understand this at this stage we will cover it and we will talk about correlational statistics]*
- **Data to collect:** screener experience item; T,C,S per participant.

Task 2. Once you have identified what data you need to collect to evaluate your hypotheses create a task that outlines the steps of the study (**session script**). This should outline what you expect the users to do and what data you will be collecting to evaluate the hypotheses you have formed.

The **test plan** (session script) should include the following information:

- The characteristics of the participants to take part at the study and how many you will recruit
- How will you split your groups (will it be within or between study?)
- A task plan to guide the users on the tasks they will have to complete
- The result that the participant will strive to achieve (refer to actual data and names rather than generalities)
- What data will you collect for each task and how will you be collecting it (you will observe the users and do not taking, you will ask a question)
- The state of the system when the task is initiated
- Readouts of displays and printouts that the participants will see while performing the task

Have a look at the example of the **session script** on Blackboard.

Model answer:

Study title

Guided Start vs Standard Flow for Creating an 8×8″ Birthday Photo Book on Snapfish

Hypotheses (two-tailed, non-directional)

- **H_0:** There is **no difference** in efficiency and success outcomes between the **Guided start** condition (pre-filtered to 8×8 birthday + inline tips) and the **Standard flow** condition.
- **H_1:** There **is a difference** in efficiency and/or success outcomes between the two conditions.
- **Outcomes (DVs): efficiency and success — i.e., time-on-task *T*, error count *E*, completion *C* (success/fail), and SEQ ease score *S* (1–7).**
- **Independent Variables (IVs):** interface variant $V \in \{guided, standard\}$

1) Participants

Target profile (screened):

- Adults in the UK, **parents** of children ≤18 years old.
- Have created **≥1 printed photo product** (any brand) in the last 24 months (yes/no).
- Mix of desktop/laptop users (Windows/macOS); Chrome/Edge/Safari up to date.
- Normal or corrected vision; comfortable with English.

Sample size & rationale: n = 24 total, **between-subjects** (12 per condition). This yields directional sensitivity to medium effects (approx. d≈0.6) but we'll analyse two-tailed.

Recruitment: University research pool + reputable online panel; screeners applied. Incentive: £15 voucher.

2) Group split & assignment

Design: Between-subjects (to avoid learning/carryover).

Randomisation: Block-random assignment by device type to balance desktop OS/browser across conditions (guided vs standard).

Blinding: Participants are naïve to variant; moderator is aware (for setup only).

3) System state before each session

- Browser cache cleared; **private window** used.
- Participant **logged out** of Snapfish; no prior cookies.
- **Assets pack** downloaded to local desktop:
  - **/BirthdayAssets/** containing **12 images**: IMG_01.jpg … IMG_12.jpg

  - **6 short captions** in captions.txt (each ≤ 60 chars).
- Snapfish UK **home page loaded**; region = United Kingdom.
- For **Guided** condition only: open the pre-filtered entry ("Photo Books → 8×8 → Birthday"), inline tips enabled (if toggle exists).
- For **Standard** condition: open **home page** (no pre-filters, no tips).

4) Moderator script (verbatim prompts)

Intro & consent (2 min)

Thanks for joining. I'll observe and take notes while you share your screen. This is about the website, not you. Feel free to think aloud. We'll stop before checkout. Is recording OK?

Warm-up (no data scored)

*Please show me where you would start if you wanted to make a **photo book**.*

Task T1 – Start the correct product (max 3 min)

*Start creating an **8×8 inch** photo book with a **birthday** theme.*

Task T2 – Add photos (max 3 min)

*Add **at least 10** photos from the **/BirthdayAssets/** folder to this project.*

Task T3 – Customise pages (max 10 min)

*Create **at least 6 inside pages**, each with a **different layout** suited to the photos you choose.*
*Add a **short caption** on **each inside page** (use captions.txt).*
*Change the **background colour** on at least **two** pages.*
*Add at least **one embellishment/sticker** somewhere.*
*Edit both the **cover** and the **back** page.*

Task T4 – Review & save (max 2 min)

*Preview your book and **save the project** named **"Liam_Birthday_8x8"**. Do **not** check out.*

Post-task mini-surveys (10–20 sec each)

*For each of the tasks we just did, please rate how easy or difficult it was on a scale from **1 (Very difficult)** to **7 (Very easy)**.*

Wrap-up (2 min)

What, if anything, felt confusing or missing while creating your book?
If you could change one thing to save time, what would it be?

5) Participant's target result (explicit, concrete)

By the end of T4, the participant should have a saved Snapfish project with:

- **Project name:** Liam_Birthday_8x8
- **Product:** Photo Book, **Size:** 8×8″
- **Theme:** any **Birthday** theme visible in the UI (e.g., “Birthday Bash” or equivalent)
- **Assets used: 10–12** images from /BirthdayAssets/ (file names recorded)
- **Pages: ≥6 inside pages** (distinct **layout IDs**), plus **cover** and **back** edited
- **Captions:** ≥6 text boxes, one per inside page (text taken from captions.txt)
- **Backgrounds:** at least **2** inside pages with non-default backgrounds
- **Embellishments:** ≥1 sticker (e.g., Balloon_Sticker_03 if named in UI)
- **Status:** Saved; preview reachable

6) Data to collect per task (with methods)

Instruments used: screen recording, observer notes, and an event checklist. (Students can emulate “event capture” with a structured tally sheet if no telemetry is available.)

Task	What the participant does	Data captured	How collected	Success criteria
T1 Start 8×8 Birthday	Navigates to Photo Books; selects 8×8; applies Birthday theme	**T (sec)**, path clicks, **E** (errors), **C** (0/1)	Timestamp start/stop; note each misclick/backtrack; mark if wrong size/theme chosen	Builder open showing **8×8** + **Birthday** theme visible
T2 Add photos	Imports ≥10 photos from /BirthdayAssets/	**T**, **C**, file names used, upload issues	Observe file chooser; note stall messages; count duplicates	Media tray shows **≥10 unique** assets added
T3 Customise pages	Builds ≥6 inside pages w/ unique layouts; captions; backgrounds; embellishments; edits cover & back	**T**, **E** (e.g., failed apply), **C**; counts: pages, **unique layout IDs**, captions/page, backgrounds changed, embellishments present	Observer checklist with tick boxes for each required element; screenshot evidence	All required customisations present (see Target result)
T4 Review & save	Previews and saves project as Liam_Birthday_8x8	**T**, **C**, save confirmation text	Note time to first preview/save; capture confirmation text	Project saved; visible in account/project area or on-page confirmation

Perception measures:

- **SEQ 1–7** after **T1, T2, T3** (ease).
- **Findability** single item 1–7 after T3 (“I could easily find the controls I needed”).
- **Post-test UMUX-Lite** (2 items, 1–7).

Observer notes template (examples):

- Misclicks/backtracks; **error toasts** text (verbatim); rage-click moments; visible hesitations (≥3s hovering).
- Detours (e.g., navigates to prints by mistake).
- Participant verbalisations indicating confusion ("I can't find backgrounds").

7) Readouts the participant will see (document verbatim text)

Students should paste real labels as they appear during piloting. Example placeholders to record:

- **Buttons/links:** "Photo Books", "8×8", "Add photos", "Themes", "Birthday", "Layouts", "Backgrounds", "Stickers/Embellishments", "Text", "Preview", "Save project".
- **System confirmations:**
  - "8×8 selected" (size chip highlighted)
  - "10 photos added" (toast or media tray count)
  - "Theme applied: Birthday Bash"
  - "Project saved: Liam_Birthday_8x8"
- **Errors (write verbatim if they appear):**
  - "Upload failed. Try again."
  - "Layout could not be applied."

These strings are copied into the session log to support coding of **E** (errors) and to verify **C** (completion).

8) Measurement & scoring rules

- **Time T:** wall-clock per task (start when the prompt is read, end when success event visible).
- **Completion C:** 1 if task success event achieved within time cap; else 0.
- **Errors E:** count of blocking issues (wrong size/theme/failed save/failed upload needing retry).
- **SEQ S:** 1–7 single item per task.

One-page observer checklist (students can print)

- Participant ID / Condition: □ Guided □ Standard
- Device / Browser: _____
- **T1** Started 8×8 Birthday — Start ***:*** End ***:*** **C:** □1 □0 **E:** ___
- **T2** Added ≥10 photos — Start ***:*** End ***:*** **C:** □1 □0 **E:** ___ Files used: _____
- **T3** Customised pages — Start ***:*** End ***:*** **C:** □1 □0 **E:** ___

- Cover edited □ Back edited □

- **T4** Preview & Save — Start **:** End **:** **C:** □1 □0 **E:** ___ Confirmation text: _____
- SEQ (T1/T2/T3): __ / __ / __
- Notes (errors verbatim / confusions): _____

Task 3. For all the questions you will use to collect data create the respective:

- **pre-test**,
- **in-test,** and
- **post-test questionnaire**

Use **Microsoft 365 Forms** to turn those questions into a form that you can distribute to users to collect data that will help you to evaluate your research question. Make sure you give the right permissions to the form for the users to complete it.

After the completion of the **Microsoft 365 Form** create a spreadsheet that will be collecting the data. You will need this to analyse the data in the following tutorials.

Each group should form at least **2 pre-test**, **6 in-test,** and **2 post-test questions**.

1) Sample Pre-test (at least 2 questions)

Purpose: capture participant traits that may influence outcomes (IVs/covariates).

1. **Which version are you using today?**
   Type: Choice (required) — "Guided start (pre-filtered + tips)", "Standard flow"
   Note: If you don't want participants to see this, prefill it or have the moderator select it.
2. **Which device are you using today?**
   Type: Choice (required) — "Desktop/Laptop", "Mobile", "Tablet"
3. **Which OS and browser are you using?**
   Type: Text (short) — example placeholder: "Windows 11 / Chrome"
4. **How many times have you made a printed photo product in the last 24 months?**
   Type: Choice (required) — "None (0)", "Once (1)", "A few times (2)", "Many times (3)"
5. **How familiar are you with Snapfish?**
   Type: Likert 1–5 — labels: 1 "Not at all" … 5 "Very familiar"

2) In-test (≥6 questions; shown during/after tasks)

These are quick items the participant answers immediately after each task in your unmoderated flow.

After Task 1 — Start 8×8 Birthday

1. **T1 SEQ:** Overall, how easy or difficult was this task?
   Type: Likert 1–7 — 1 "Very difficult" … 7 "Very easy"
2. **T1 Completion:** Did you complete the task successfully?
   Type: Choice — "Yes", "No"

After Task 2 — Add ≥10 photos

3. **T2 SEQ:** Overall, how easy or difficult was this task?
   Type: Likert 1–7
4. **T2 Completion:** Did you complete the task successfully?
   Type: Choice — "Yes", "No"

After Task 3 — Customise pages

5. **T3 SEQ:** Overall, how easy or difficult was this task?
   Type: Likert 1–7
6. **Findability:** I could easily find the controls I needed.
   Type: Likert 1–7 — 1 "Strongly disagree" … 7 "Strongly agree"
7. **Mental Effort:** How mentally demanding was this overall?
   Type: Likert 1–7 — 1 "Not at all" … 7 "Extremely"
8. **Did you notice any tips or guidance while building the book?**
   Type: Choice — "Yes", "No", "Not sure"

3) Post-test (≥2 questions)

1. **UMUX-Lite — The system's capabilities meet my requirements.**
   Type: Likert 1–7 — 1 "Strongly disagree" … 7 "Strongly agree"
2. **UMUX-Lite — The system is easy to use.**
   Type: Likert 1–7 — 1 "Strongly disagree" … 7 "Strongly agree"
3. **What, if anything, made any task harder than expected?**
   Type: Long text (optional but recommended)
4. **If you could change one thing to save time, what would it be?**
   Type: Long text (optional but recommended)

Task 4. After you have completed your Task plan and questionnaire share it on this Padlet to allow users to complete the task and provide you with the required data.

Task 5. Access the Ethics forms that need to be completed before you conduct a study involving participants at the University of Westminster available on Blackboard. Look at a completed example. How those forms will be modified for your study?

Task 6. Experiment with a tool like https://lookback.io/ Maze/UserTesting/Useberry which is designed for online usability testing sessions, create a session for your users, share the link to participants, record a session with them and playback your recording. Perhaps you could consider this tool to collect data for your Coursework or use **Microsoft Teams** that allows to record a session and keep a transcript of a session.